



Ceren Çatak

Phone: (+90) 5462630913 (Mobile) | **Email address:** cerencatak00@gmail.com | **LinkedIn:** www.linkedin.com/in/ceren-catak | **GitHub:** <https://github.com/cerencatak> | **Address:** Yaşamkent, Çankaya, Ankara, Türkiye (Home)

● ABOUT MYSELF

Geomatics Engineering | Geospatial Data Processing | Web GIS | Remote Sensing | Photogrammetry

4th-year Geomatics Engineering student focused on geospatial data production, mapping systems, satellite-image-based analysis and Web GIS development. Experienced in UAV data collection, photogrammetry, PostGIS workflows and remote sensing applications. Strong interest in image processing, data analysis, real time mapping/localization and defense-industry geospatial technologies.

● EDUCATION & TRAINING

05/09/2021 - Current - ANKARA, TÜRKİYE

GEOMATICS ENGINEERING- HACETTEPE UNIVERSITY

Final grade: 2.52 / 4 | **Level in EQF:** 6

● WORK EXPERIENCE

07/07/2025 - 02/08/2025 - ANKARA, TÜRKİYE

INTERNBMD SURVEYING ENGINEERING CONSULTANCY

Conducted high-precision photogrammetry and stereo mapping workflows for engineering-oriented mapping projects. Applied PPK-related production concepts to support positional accuracy and quality-controlled data outputs. Performed technical checks during final product preparation, improving mapping-standard awareness and spatial data reliability.

07/07/2024 - 05/08/2024 - ANKARA, TÜRKİYE

INTERNGENERAL DIRECTORATE OF TURKISH COAL ENTERPRISES

Managed and processed large-scale mining license data using Netcad, supporting regulatory and technical data integrity. Performed UAV-based field measurements and data collection for topographic mapping and spatial documentation. Collaborated with engineering teams to streamline spatial analysis workflows and field-to-office data production processes.

● SKILLS

Programming & Data

Python | Java | MATLAB | SQL | PostgreSQL | PSQL-PostGIS | basic algorithm development | data analysis

GIS & Remote Sensing

QGIS | ArcGIS | Google Earth Engine | satellite image processing | NDWI/MNDWI | geospatial analysis

Web GIS & Development

Leaflet | JavaScript | HTML/CSS | FastAPI basics | REST API workflow | Git/GitHub | DevOps

Mapping & Surveying

GNSS data processing | coordinate systems | geodesy | UAV operations | photogrammetry | stereo mapping

Production Tools & Interests

AutoCAD | NetCAD | Agisoft Metashape | computer vision | AI-based image analysis | sensor data interpretation

● PROJECTS

06/10/2025 - 24/04/2026

Turkey Lakes Monitoring Platform - Graduation Project | Web GIS, Remote Sensing, Google Earth Engine

Developed a Web GIS monitoring platform to analyze long-term lake surface changes in Türkiye using Landsat-5/7/8 and Sentinel-2 satellite imagery. Processed data in Google Earth Engine and applied NDWI/MNDWI indices for water extraction, temporal comparison and spatial interpretation. Built interactive map/dashboard components with time-slider visualization, shrinkage trends, risk levels and climate-variable interpretation.

Link <https://turkiyelake.com>

22/02/2026 - CURRENT

FIG Web GIS Project – Full-Stack Web GIS Application

Currently contributing to an ongoing full-stack Web GIS project involving spatial data management, route analysis, backend API services, and interactive map-based visualization. Took part especially in the deployment stage, including server-side setup, build processes, Nginx configuration, application publishing, and testing the live environment. The project improved practical knowledge of Web GIS deployment workflows, PostgreSQL/PostGIS integration, FastAPI-based services, and production-oriented geospatial software development.

02/03/2026 - 07/05/2026

Omni Monitor – Web GIS Monitoring and Visualization Project

Worked on a Web GIS-based monitoring and visualization project focused on presenting real-time spatial data through interactive maps, analytical panels, and user-oriented dashboard components. The project strengthened practical experience in map-based data visualization, frontend GIS interfaces, spatial analysis workflows, and the interpretation of geographic information for monitoring and decision-support purposes.

HidroRisk - Micro-Scale Flood Risk Analysis (Full-Stack Web GIS Application)

Developed a Web GIS application for spatial risk analysis using PostgreSQL/PostGIS, Python-based tools and interactive map visualization. Implemented spatial database workflows, geospatial queries, map layers and API-oriented data transfer for location-based risk information. Used Git/GitLab and DevOps-oriented teamwork practices for collaborative development, version control and deployment-oriented workflows.

● CONFERENCES & SEMINARS

01/02/2026 - 07/02/2026

HKMO - Software Development Camp

Developed a full-stack Web GIS application for micro-scale flood risk analysis.

- Implemented spatial data management and complex queries using PostgreSQL/PostGIS.
- Built dynamic map visualizations using Python-based development tools and modern web technologies.
- Managed collaborative workflows using Git for version control and followed DevOps principles.

● CERTIFICATIONS

09/02/2024 GENERAL DIRECTORATE OF CIVIL AVIATION

UAV Pilot Certificate (IHA-1)

Qualified for high-precision aerial data collection and autonomous flight operations

12/02/2026 KOSGEB

KOSGEB Entrepreneurship Training Certificate

● **DRIVING LICENCE**

Driving licence: B | 23/08/2024 - 23/08/2034

● **LANGUAGE SKILLS**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	B2	B2	B2	B2	B2
GERMAN	A2	A2	A1	A1	A1